

Six-Month Project Synopsis

Project Title

Vision-Based Autonomous Pick-and-Place Robotic Arm Using Imitation Learning and Digital Twin

Project Overview

The objective of this project is to transform a 6-DOF robotic arm into an autonomous vision-guided manipulation system inspired by modern embodied AI frameworks such as LeRobot. Rather than relying on manually programmed robotic movements or traditional Inverse Kinematics for autonomous execution, the proposed system will learn pick-and-place behaviour directly from human demonstrations collected through teleoperation.

The robotic arm is a serial manipulator fabricated using **three MG996R servo motors** and **three MG90S servo motors** (including the gripper), providing true 6 Degrees of Freedom. The focus of this project is to develop an imitation learning pipeline capable of learning complex robot motion from demonstrations.

During the demonstration phase, the robotic arm will be controlled using a **PlayStation 5 (PS5) controller** connected to a MacBook. To ensure smooth, high-quality demonstration data, an **Inverse Kinematics (IK) solver** will map the human's intuitive Cartesian commands (X, Y, Z) to the 6 independent joint angles in real-time. Two fixed USB cameras will continuously observe the workspace while these robot joint commands are simultaneously recorded.

Instead of providing raw images directly to the learning model, a computer vision module developed using OpenCV will first extract relevant scene information. To guarantee perception robustness against variable lighting, **ArUco markers** will be utilized during the initial validation phase. The extracted features will be converted into **calibrated real-world coordinates** (cm from the robot base) before being fed to the neural network. This hybrid approach significantly reduces the complexity of the learning problem and improves generalization while maintaining the advantages of learning-based manipulation.

The complete system will initially be developed and trained on a MacBook and later deployed on a **Raspberry Pi 5** for standalone autonomous operation. In parallel, a digital twin of the robotic arm and workspace will be developed for simulation, visualization, and future algorithm validation.

Project Objectives

- Develop an imitation learning-based robotic manipulation system.
- Develop a PS5 controller-based teleoperation system utilizing Inverse Kinematics

for smooth demonstration collection.

- Build a synchronized dual-camera data acquisition framework.
 - Develop a robust OpenCV computer vision module for workspace perception (utilizing ArUco markers and color spaces).
 - Train an imitation learning model capable of generating full 6-DOF robotic arm trajectories.
 - Develop a digital twin of the robotic arm and workspace.
 - Deploy the trained model on a Raspberry Pi 5.
 - Evaluate the system under progressively increasing task complexity.
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System Architecture

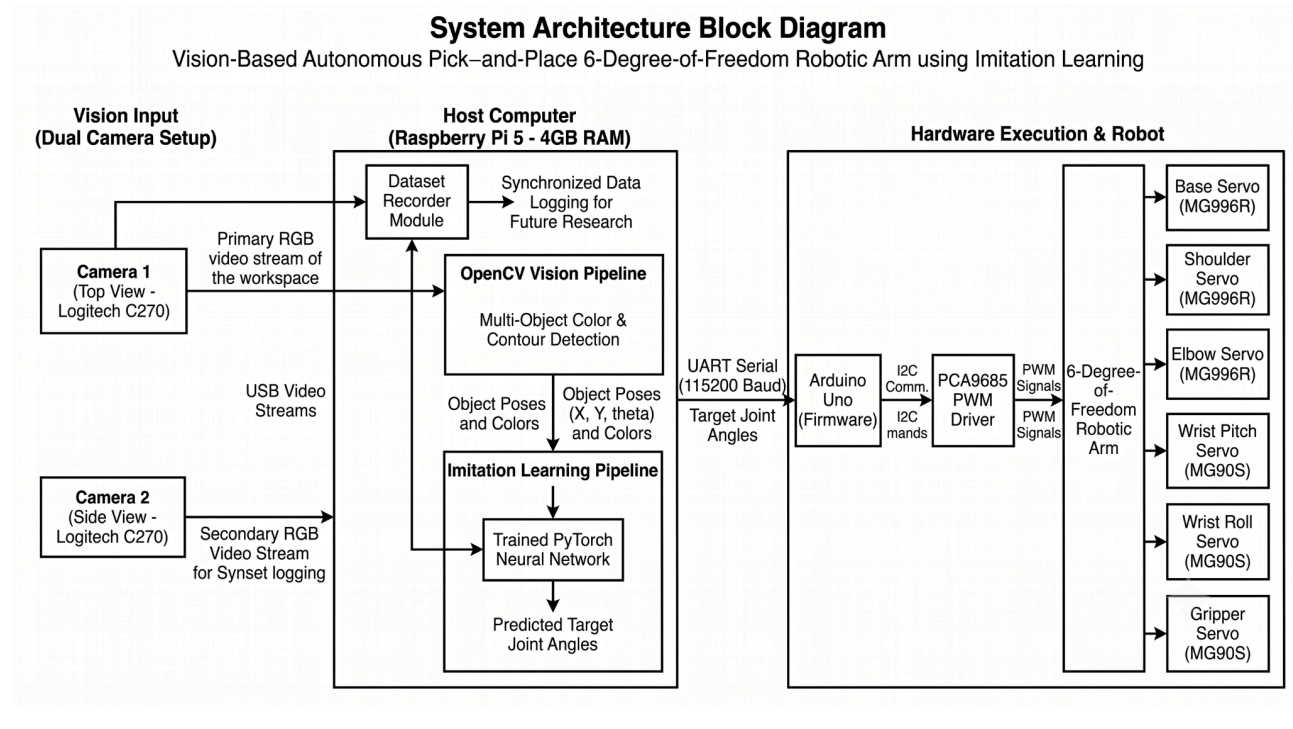
Hardware

- 6-DOF Serial Robotic Arm
- 3 × MG996R Servo Motors
- 3 × MG90S Servo Motors (including gripper)
- Arduino Uno (Microcontroller for low-level PWM driver communication)
- PCA9685 16-Channel PWM Servo Driver
- Raspberry Pi 5 (Deployment)
- MacBook (Training & Dataset Collection)
- Two Fixed USB Web Cameras (Camera 1 – Top View, Camera 2 – Side View)
- PlayStation 5 Controller (Teleoperation)
- Physical Emergency Stop Switch

Workspace

- Approximately 40 cm × 40 cm
- Robot mounted at one end of the workspace.
- Cameras remain fixed throughout the project.
- Robot always starts from a predefined, mechanically optimized **Home Position**.
- Robot always returns to the Home Position after completing each task to standardize the learning episodes.

System Block Diagram

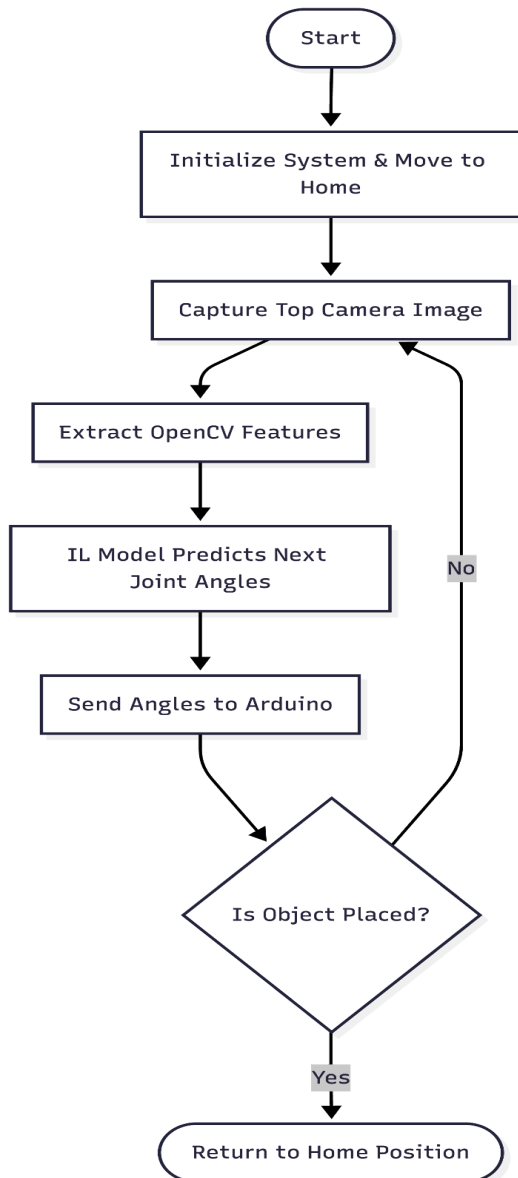


Methodology

The project follows an imitation learning approach. During each demonstration:

1. The operator controls the robotic arm's end-effector in Cartesian space using the PS5 controller.
2. The Python host calculates the required joint angles via Inverse Kinematics.
3. The robot starts from the predefined Home Position.
4. Both cameras capture synchronized RGB images.
5. OpenCV extracts scene information (e.g., ArUco poses, color centroids) and maps them to real-world coordinates.
6. Robot joint commands and gripper actions are recorded simultaneously.
7. The complete demonstration is stored as a synchronized training episode.
8. The robot returns to the Home Position before the next demonstration.

System Workflow



Learning Pipeline

Vision Module

The dual-camera vision system processes the captured images using OpenCV. The learning model **does not** receive raw images directly.

Stage 1 (Baseline) Extracted Inputs:

- Block World Position (X,Y) and Orientation (θ) via ArUco markers
- Box World Position (X,Y) and Orientation (θ) via ArUco markers

Stage 2 (Complex) Extracted Inputs:

- Block 1 & 2 World Position (X,Y), Orientation (θ), and Colour
- Box 1 & 2 World Position (X,Y), Orientation (θ), and Colour

Imitation Learning Model

The learning model receives the processed physical coordinates from the vision module and predicts the robotic arm's next joint state trajectory.

Output: A sequence of 6 joint angle commands representing the complete pick-and-place trajectory back to the Home Position.

Experimental Stages

Stage 1 – Baseline Learning (System Validation)

- **Environment:** One coloured block, One target box. The **box remains fixed**, while the **block is placed randomly**.
- **Perception:** ArUco markers used to guarantee 100% robust pose estimation regardless of lighting.
- **Objective:** To validate the complete teleoperation, Inverse Kinematics, and imitation learning pipeline by autonomously picking a randomly positioned object and placing it into a fixed destination.
- **Expected Demonstrations:** 150–250.

Stage 2 – Complex Learning (Multi-Object Sorting)

- **Environment:** Two coloured blocks, Two corresponding coloured target boxes. All objects are placed randomly.
 - **Perception:** Transitioning to raw color/contour detection as the ML pipeline is now validated.
 - **Objective:** To enable the robot to identify the correct object, associate it with the matching destination based on colour, and perform autonomous operations under increased environmental complexity.
 - **Expected Demonstrations:** 250–500.
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Work Distribution

Team Member – Soham Bhavsar

Responsibilities:

- Development of PS5 controller teleoperation software incorporating Inverse Kinematics (IK).
- Integration and calibration of 6-DOF robotic arm hardware.
- Camera integration and synchronization.
- Development of OpenCV vision module (implementing ArUco marker detection and

world-coordinate mapping).

- Development of synchronized dataset recording software.
- Development of imitation learning pipeline infrastructure.
- Raspberry Pi 5 deployment.
- System integration, testing, debugging, and optimization.
- Documentation and final report preparation.

Team Member – Divyansh Dewangan

Responsibilities:

- Development of demonstration collection methodology.
- Collection of synchronized human demonstrations.
- Generation and preprocessing of Stage 1 and Stage 2 datasets.
- Training and fine-tuning of the Stage 1 imitation learning model.
- Training and fine-tuning of the Stage 2 imitation learning model.
- Performance comparison and experimental evaluation under varying complexities.
- Documentation of experimental results.

Team Member – Toshal Kumbhar

Responsibilities:

- Development of complete CAD model for the true 6-DOF serial arm.
- Development of the digital twin and workspace modelling.
- Camera placement within the simulation environment.
- Motion simulation and validation of digital twin trajectories.
- Assistance with simulation-to-real (sim-to-real) integration.
- Documentation of the simulation framework.

Six-Month Work Plan

Month 1 – Hardware and Teleoperation Development

- Finalize hardware architecture and mechanically optimize joint limits.
- Integrate robotic arm with Arduino Uno and calibrate PWM drivers.
- Develop Inverse Kinematics solver and PS5 controller interface.
- Install dual-camera system.
- Define physical Home Position and motion constraints.

- Begin digital twin development.

Month 2 – Vision System and Dataset Framework

- Develop OpenCV perception module (ArUco tracking for Stage 1).
- Detect object position and orientation in real-world coordinates.
- Synchronize camera feeds with teleoperation loop.
- Develop dataset recording framework.
- Validate synchronized recording integrity.
- Continue digital twin development.

Month 3 – Stage 1 Dataset Collection

- Collect approximately 150–250 baseline demonstrations.
- Organize and preprocess dataset.
- Validate trajectory recordings.
- Complete first version of digital twin.

Month 4 – Stage 1 Model Training and Stage 2 Dataset Collection

- Train Stage 1 imitation learning model.
- Validate autonomous baseline operation.
- Implement color-based OpenCV tracking for Stage 2.
- Collect Stage 2 complex demonstrations.
- Record approximately 250–500 complex demonstrations.

Month 5 – Stage 2 Model Training and System Integration

- Train Stage 2 imitation learning model.
- Deploy trained models on Raspberry Pi 5.
- Integrate digital twin for visualization.
- Optimize overall system latency and performance.
- Perform autonomous physical testing.

Month 6 – Evaluation and Documentation

- Evaluate Stage 1 and Stage 2 manipulation accuracy.
- Compare system performance under increasing task complexity.
- Finalize digital twin.
- Complete technical documentation.
- Prepare final report, presentation, and capstone project demonstration.

Expected Outcomes

At the completion of the project, the following deliverables are expected:

1. A fully autonomous 6-DOF robotic arm capable of performing vision-guided pick-and-place tasks using imitation learning.
2. A synchronized dual-camera demonstration dataset collected through an IK-assisted PS5 controller teleoperation framework.
3. An OpenCV-based perception module capable of extracting object poses and mapping them to real-world coordinates.
4. Trained Imitation Learning models for both Stage 1 (Baseline) and Stage 2 (Complex) environments.
5. A complete digital twin of the robotic arm and workspace.
6. Raspberry Pi 5 deployment for standalone autonomous operation.
7. Complete technical documentation, experimental results, and capstone project report.